

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_random_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_161549_b
- started_at_local: 2026-05-13 16:15:49
- finished_at_local: 2026-05-13 16:26:40
- duration_hhmm: 00:11
- duration_seconds: 650.718
- seed_offset: 1000
- replicate_label: b
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.242634	0.036251	0.139442	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.189005	0.008454	0.098729	0.64691	0.58	0.010472
atsp_failure_replay	failure	score_only	False	0.193498	0.009004	0.101251	0.709445	0.58	0.008788
atsp_failure_replay_compression	failure	novelty_gate	True	0.255673	0.036251	0.145962	0.0	0.78	0.0

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.242634, synthetic 0.036251, combined 0.139442.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.242634 across 4 instances; family means: ft=0.4, ftv=0.376317, p=0.013523, ry=0.180696.
- Panel synthetic_holdout mean gap 0.036251 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.059392, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 2074, "family": "ftv", "name": "ftv38", "optimality_gap": 0.355556}, {"best_known_cost": 1473, "cost": 1975, "family": "ftv", "name": "ftv35", "optimality_gap": 0.340801}, {"best_known_cost": 1286, "cost": 1642, "family": "ftv", "name": "ftv33", "optimality_gap": 0.276827}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.189005, synthetic 0.008454, combined 0.098729.
- Accepted-epoch count 6, mean accepted code novelty 0.64691, and final complexity 0.58.
- Adaptation efficiency 0.010472 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.189005 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004982, ry=0.093884.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.193498, synthetic 0.009004, combined 0.101251.
- Accepted-epoch count 6, mean accepted code novelty 0.709445, and final complexity 0.58.
- Adaptation efficiency 0.008788 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193498 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.255673, synthetic 0.036251, combined 0.145962.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.255673 across 4 instances; family means: ft=0.426937, ftv=0.398016, p=0.022242, ry=0.175496.
- Panel synthetic_holdout mean gap 0.036251 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.059392, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1911, "family": "ftv", "name": "ftv35", "optimality_gap": 0.297352}, {"best_known_cost": 1286, "cost": 1570, "family": "ftv", "name": "ftv33", "optimality_gap": 0.22084}, {"best_known_cost": 1530, "cost": 1857, "family": "ftv", "name": "ftv38", "optimality_gap": 0.213725}]

Judge Appendix

Benchmark Suite Analysis: Replay Modes on ATSP

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Complexity	Replay Mode	Notes
atsp_no_replay	0.242634	0.036251	0.139442	0.0	0.78	none	Baseline with no replay; lowest transfer performance
atsp_random_replay	**0.189005**	**0.008454**	**0.098729**	0.64691	0.58	random	Best overall transfer and optimality gaps; higher code novelty but simpler solutions
atsp_failure_replay	0.193498	0.009004	0.101251	0.709445	0.58	failure	Slightly worse transfer than random replay, but still improved over no replay with high code novelty
atsp_failure_replay_compression	0.255673	0.036251	0.145962	0.0	0.78	failure + compression	Worst transfer and optimality gaps; zero code novelty; highest complexity

Key Interpretations

- ****Transfer Evidence (TSPLIB & Synthetic Gaps):****
- Random replay condition (atsp_random_replay) clearly outperforms no replay and failure-based replay modes in both held-out TSPLIB and synthetic holdout gaps (TSPLIB gap 0.189 vs. 0.243 no replay and ~0.193 failure replay).
- Synthetic holdout gaps are lowest under random replay (0.0085), indicating strong generalization to synthetic unseen problem structures.
- atsp_failure_replay_compression shows degradation in transfer metrics, indicating compression pressure under failure replay negatively impacts generalization.
- ****Code Novelty vs Transfer:****
- Random and failure replay show substantial mean code novelty (~0.65-0.71), contrasted with no novelty for no replay and failure replay compression.
- Despite higher code novelty under random and failure replay, complexity is lower (0.58 vs 0.78), which suggests replay aids in discovering simpler yet more transferable solutions.
- The failure replay compression condition loses code novelty without improving transfer, indicating code novelty reduction corresponds with a transfer decline in this mode.

- **Replay Mode Distinctions:**
- **No Replay:** Baseline performance with moderate complexity and no code novelty.
- **Random Replay:** Best transfer metrics and gaps, high code novelty, and simpler policies, showing replay of random past experiences aids transfer.
- **Failure Replay:** Slightly weaker transfer than random replay but still better than no replay; highest code novelty.
- **Failure Replay + Compression:** Compression pressure results in no code novelty and worst transfer, indicating negative effects from over-compression despite failure replay.

Summary

- The **random replay** condition achieves the strongest transfer performance on held-out TSPLIB and synthetic ATSP benchmarks, with the smallest optimality gaps by a notable margin.
- This improvement in transfer occurs despite a reduction in final solution complexity compared to no replay, indicating effective learning of generalizable and simpler strategies.
- Code novelty is significantly higher under replay conditions, particularly random and failure replay modes, but this novelty correlates with improved or stable transfer—except under compression where novelty and transfer both degrade.
- Failure replay without compression provides marginally inferior transfer compared to random replay but considerably better than no replay.
- Compression combined with failure replay negatively impacts both code novelty and transfer, highlighting a tradeoff between compression pressure and generalization.

Therefore, **random replay mode** is identified as the best replay strategy, improving generalization and optimality gaps over both no replay and failure replay modes, even though code novelty fluctuates. Compression-aware replay reduces code novelty and transfer, arguing against its use in this benchmark setting.